

Daman Dhaliwal

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Technical Skills

Programming & Data Systems: Python, SQL, R, Julia, TypeScript, Bash, Git, Docker, REST APIs, pytest, Power BI
Causal Inference & Statistical Modeling: statsmodels, Panel Data Methods, Double/Debiased ML, Causal Forests, IV, Fixed Effects, Sensitivity Analysis, Experimental Design, Heterogeneous Treatment Effects
Artificial Intelligence & Machine Learning: scikit-learn, Gaussian Processes, Variational Autoencoders, Diffusion Models, Transformer Architecture, Model Evaluation, LLM-Assisted Extraction, Gemini API, Browserbase Stagehand

Experience

Senior Financial Analyst, Chard | Vancouver, BC June 2022 – July 2025

- Supported **\$1.0B** in institutional capital allocation across **6 projects** by building analytical models to forecast asset performance, stress-test downside scenarios, and translate market, cost, rent, and financing assumptions into investment decisions.
- Built an AI-assisted market intelligence pipeline using **Browserbase Stagehand**, **Gemini API**, **TypeScript**, and **Python** to extract municipal filings, structure unstructured data, and combine outputs with market datasets, reducing research turnaround by **50%**.
- Converted spreadsheet-based forecasting workflows into **MS SQL** analytical systems connected to Sage accounting data, enabling repeatable model inputs, variance diagnostics, scenario testing, and executive-facing performance monitoring.
- Implemented **data contracts** across project financial models, standardizing inputs, assumptions, metric definitions, forecast updates, and reporting outputs into a consistent source of truth for finance, development, and executive stakeholders.
- Developed **Power BI** dashboards tracking **11 projects** and partnered with project managers on monthly and quarterly health reviews; briefed the **CEO, COO, and CFO** on performance drivers, emerging risks, forecast changes, and recommended actions.

Empirical Researcher, Journal of Political Economy, University of Chicago | Chicago, IL July 2025 – Present

- Audited empirical research codebases by executing author-provided pipelines across **R, Julia, Python, Stata, and shell**, validating statistical methods, generated outputs, and causal-inference claims against manuscripts.
- Co-developed standardized replication workflows with **Bash, Git, Docker, and virtual machines**; documented method, dependency, and data-quality discrepancies in review-ready reports on reproducibility and empirical credibility.

Data Scientist, Sciences Po | Paris, France January 2024 – December 2024

- Engineered a **real-time geospatial data pipeline** integrating **UK Government Transit APIs** and the **Open Source Routing Machine (OSRM)** to ingest, process, and store **50M+** GPS observations, enabling large-scale performance benchmarking across London's bus network.
- Built route-level efficiency and utilization measures by classifying vehicle states, separating revenue service from dead miles and parked time, and validating outputs against scheduled route patterns.

Selected Projects

PyFixest (1M+ Downloads) | Core Contributor, py-econometrics github.com/py-econometrics/pyfixest

- Implemented configurable demeaning iterations, **Driscoll–Kraay** and **Newey–West** standard errors, and sensitivity analysis in a production-grade open-source library; delivered end-to-end feature development across **API design**, numerical correctness, backward compatibility, tests, documentation, and edge cases.

Regime-Switching GP Asset Pricing github.com/damandhaliwal/RSGP

- Built a point-in-time ML forecasting pipeline converting raw financial, firm-level, macro, and volatility data into **Parquet panels** and evaluation tables, then implemented a **Gaussian Process** model with non-stationary **hidden Markov chain** transition probabilities and out-of-sample validation.

Viral Voting: Information Diffusion Effects of Social Pressure github.com/damandhaliwal/ViralVoting

- Took an existing 344K-observation field-experiment dataset and applied **X-Learner** and **Causal Forest DML** to estimate heterogeneous treatment effects, finding nonlinear social-pressure effects with diffusion gains that attenuate at higher exposure levels.

Education

MA, Economics (Research Emphasis) | University of Toronto May 2026

Coursework: Applied Causal Machine Learning (A+), Statistical Methods for Machine Learning and Data Mining (A-), Econometrics (A), Economics of Algorithms (A+), Computational Methods for Heterogeneous-Agent Models (A+)

B.Comm, Finance (with Honours) | University of British Columbia May 2022

Award: Karen McKellin International Leader of Tomorrow Award (\$300,000 merit-based)